Black Box or Sandbox: Computational Approaches to Interpreting Michigan's Clean Slate Law

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Abstract

The U.S. State of Michigan's Clean Slate Statute, defining conditions under which prior convictions can be removed from an individual's public record, has allowed thousands of people to lead freer and more productive lives. Yet the complex and evolving nature of the law has led to confusion, disagreement, and uncertainty. The Clean Slate laws went into effect in 2021 allowing individuals to apply for an expungement from the court in which they were convicted. The law also required the Michigan State Police to find a way to automate this process within two years. In compliance with that mandate, the Michigan State Police have implemented and deployed a "Rules Engine" in 2023 that automatically detects some expungeable convictions and removes them from individuals' records. The Rules Engine has greatly increased the throughput of set-aside cases and scaled up the benefits of the Clean Slate laws, but in its current form it is a black-box solution with minimal explanation for the general public. There is a risk that this automation will diminish individuals' sense of agency and engender a passive attitude toward their criminal records.

Our contributions in this paper are threefold. First, we articulate some of the confusion and disagreement over the interpretation of the Clean Slate statute, through interviews with legal professionals. Second, we explain the operation of the Rules Engine, following the relatively sparse explanation available, and we discuss the consequences of this effective but inscrutable tool. Finally, we present our initial work on an alternative computational approach: a "digital sandbox" for legislators, lawyers, judges, and citizens to define and explore the consequences of legislation. The sandbox environment will allow differing interpretations of statutory law, or various revisions to existing law, to be expressed formally and compared through automated analysis. We see computation as a way not only to expand the benefits of criminal justice reform but also to enhance individuals' understanding and agency with regard to the law.

1 Introduction

The notoriously complex and specialized text and structure of statutory legislation can generate confusion and disagreement in interpretation even among experts and can leave ordinary citizens disempowered. Compounding the problem, the dynamic nature of legislation can make it difficult for stakeholders to anticipate the consequences of proposed amendments. This can have a chilling effect on laws clearly intended to help citizens — such as the State of Michigan's Clean Slate statute. (Michigan Legislature 2021 hereinafter "MCL 780.621"). Our team of computer scientists and lawyers has been examining this consequential, complex, and evolving piece of legislation, which establishes conditions under which prior convictions can be removed from an individual's public record. Two of our team members have in fact been helping individuals apply for conviction set-asides, alternatively referred to as expungements. We have seen first-hand what a tremendous positive effect a clean record can have in people's lives, but interviews conducted with other legal experts who regularly deal with the statute have elicited difficulties they face with the statute's language: differences in interpretation, problems understanding the consequences, and a general sense of uncertainty that limits the full purpose of the statute.

In this paper, we present some of the experiences of legal professionals who work with the statute, and we illustrate some of the differing interpretations taken by lawyers and judges. We then discuss the operation of the "Rules Engine", an automated tool deployed in 2023 by the Michigan State Police to detect and expunge qualifying convictions. The appearance of the Rules Engine has vastly increased the scale of the expungement project in Michigan, but its black-box nature and somewhat idiosyncratic behavior may have negative effects on individuals' sense of empowerment. As an alternative example of bringing computational power to bear on this problem, we discuss our preliminary work on developing a "digital sandbox" for professionals and ordinary citizens to explore the consequences of legislation.

2 Michigan's Clean Slate Act

A criminal record, no matter how distant, creates barriers to employment, housing, and benefits. These barriers, which have been described as amounting to a "new civil death," magnify social, economic, and racial disparities [1]. A core goal of this century's emergent criminal justice reform movement in the United States has been to lessen barriers faced by people with criminal records, and a key tool has been legal mechanisms to expunge or set aside old criminal convictions. As of 2023, almost all U.S. states have some form of record relief, and 32 states plus the District of Columbia have authorized automatic expungement of criminal records [14].

Through its 2021 Clean Slate Act (MCL 780.621 *et seq.*), the State of Michigan is at the forefront of this movement. As with other states, criminal records do not entirely disappear, as the State Police maintain a non-public record of the conviction that will be available if there is a new interaction with courts. However, records that have been set aside are not available to landlords or employers during background checks. The Michigan State Court's website [18] cites a University of Michigan study finding that those whose criminal records are set aside experience "a sharp upturn in their wage and employment trajectories" [24]. This view is certainly shared by the thousands of individuals who have received expungement relief since the new rules came into effect in April 2021. Through the work of a dedicated group of legal aid agencies and citizens' groups it appears that out of 2.8

How confident on a scale of 1-5 (1 being not at all confident and 5 being extremely confident) are you that:	Respondent				
	A	B	C	D	E
(1) the statute is interpreted and understood in a unified way by legal professionals across the state?	4	3	2	3	1
(2) legal professionals understand the consequences of the statute?	5	4	5	4	4
(3) the statute is interpreted and understood in a unified way by judges across the state?	4	3	2	1	1
(4) ordinary citizens interpret the statute correctly and understand its consequences?	2	2	2	1	1

Table 1: Respondents' answers to survey questions.

million people with criminal records in Michigan almost 280,000 individuals have been fully freed from the burden of a criminal record through the Clean Slate Act as of September 2023 [30].

In drafting the Clean Slate Act, the Michigan Legislature set limits on which crimes and how many crimes can be expunged, and on how long an individual must wait after a conviction to become eligible. This was not a simple calculus, as legislators and citizens have conflicting and passionate opinions on all of these factors, and the resultant legislation is a complex document that reflects many compromises. The language has been subject to varied interpretations by attorneys and the judges tasked with ruling on these cases. Moreover, it is an evolving document as there have also been amendments to the law. For instance, the 2021 law initially excluded set-asides for operating a vehicle while intoxicated (OWI), but about a year later, in February 2022, the statute was amended to allow one OWI conviction to be set aside. See MCL 780.621c(3). Other amendments are likely to come in the near future as legal aid and citizen groups work to lobby for differing protections.

3 Stakeholder experiences and perspectives

Legal professionals who work on expungement cases testify to the confusion created by the statute's multiple interpretations and the inconsistency surrounding judicial interpretation of the set aside rules. Our project team interviewed a number of these professionals, focusing on the following questions:

- To what degree is the statute, and its consequences, interpreted or understood in a unified way by judges and legal professionals across the state?
- To what degree do ordinary citizens interpret the statute correctly and understand its consequences?

We asked a small sample of attorneys to answer some Likert-scale questions and follow up with free comments on their experiences with the Clean Slate Act. Our respondents are all attorneys with significant experience assisting clients with set-aside applications, ranging from 100 to 500 such cases.

Our interviews started with a series of Likert-scale questions, shown in Table 1. Respondents generally felt confident that the *consequences* of the statute are understood by legal professionals in a unified way. However, they generally indicated low confidence in the unity of *interpretation*

of the statute by both legal professionals and judges. They also overwhelmingly reported low confidence in the understanding of the statute by ordinary citizens.

Consistently with the Likert scale results, the open-ended responses suggest that ordinary Michigan citizens have a limited understanding of the statute. This issue is particularly salient with those who seek expungements. Respondent B recalled their experience at set-aside clinics at fairs in which "[p]eople will come to us... and say 'I want to get my... [conviction] set aside so that if I get in trouble again, they'll treat it like a first offense'". Respondents indicated that those seeking expungements are not well served by official instructions, and that they rely primarily on unreliable sources for information. As Respondent D indicated, reading complicated legal documents is a barrier that causes many to give up or not even begin the process:

> [The statute's] language is very confusing... It's huge. I have to read it nearly every day... [and] I'm constantly learning new things. I find that it's very, very difficult... for someone who has a felony conviction to figure out even what they need to do, and the instructions attached to the application are not that clear even if you were to follow them exactly.

4 The Statute and Judicial Interpretation

The wide variety of real-world cases that fall under the purview of the Clean Slate Act highlight the multiple ways in which the text of the law can be, and are, interpreted by lawmakers, judges, and legal professionals. We point out two areas that are subject to multiple interpretations: the first concerning waiting periods, and the second concerning how convictions are counted.

Waiting periods: counting forward, counting backward. The statute mandates that an application can be filed only after all prison and parole conditions have been completed, plus a seven-year wait for records with more than one felony, a five-year wait for records with just one felony, or a serious misdemeanor or Operating While Intoxicated (OWI) charge, or a three-year wait to file for most other misdemeanor convictions. This section then goes on to state that for a judge to grant the set aside, it must be shown that the "the applicant has not been convicted of any criminal offense during the *applicable time period required.*" (MCL 780.621d (4)(c), emphasis added). But the statute does not specify exactly what the applicable time period is. Should it be the date of the offense looking forward three, five, or seven years, should it be the current date looking back three, five, or seven years — or perhaps both conditions need to be met?

Our interviewees described ambiguity in the statute's language and differing interpretations by legal professionals and judges across the state on waiting periods for expungements:

> Say you have a felony or two felonies which will require a seven-year waiting period, there is one school of thought that says, hey, if you are trying to get that felony set aside and seven years have passed since the later of those two felonies, then you are good to go on that application... The other school of thought says, if you have committed crimes in the meantime, you should wait until the waiting period for that particular crime [is up].

Another interviewee experienced an even more conservative interpretation of waiting periods:

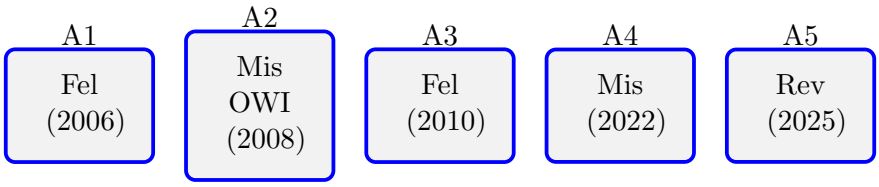

Figure 1: Expungement scenario A.

> The prosecutor argued that since the former convict had a felony in 2000 and another one two years later [in 2002]... he didn't wait seven years [between felonies]. And so, the prosecutor's interpretation was that you had to wait... five to seven years between each felony to get any of them expunged.

Interviewees also described issues with interpreting the special rules for expunging OWI (operating while intoxicated) offenses:

> That section, [which] became law in March of 2022... doesn't say that an individual is not eligible [for getting a single OWI expunged] if they've had more than one offense. But the office of the Attorney General has been opposing situations in cases where somebody has had a first time OWI, followed by a subsequent OWI.

Scenario A in Fig. 1 demonstrates how differing interpretations of this "applicable time period" rule might impact an application. The court's decision as to what offenses to set aside depends on how the particular judge reads the statute.

Interpretative approach 1 (counting *forward* from the date of conviction) is what attorneys view as the narrowest interpretation of the applicable time period provision. With this approach, the Court looks forward from the date of the conviction release to see if there is a subsequent conviction within three years (for a misdemeanor conviction), five years (for a single felony, OWI, or serious misdemeanor) or seven years (for two or more felonies) to determine if the prior conviction can be expunged.

In our sample scenario, counting forward would allow the 2022 conviction to be expunged in 2025. The 2010 conviction could be expunged in 2025 as well, but when considering the 2008 and 2006 convictions, the judge would conclude they can never be expunged because of the fact that there are intervening convictions within the "applicable time periods."

Most offenders commit their offenses within the span of a few years before their mid-20's [8, 23], and the counting forward approach presents obstacles to expungement in such cases. An individual who commits three felonies eight years apart (over a 24 year period) may have them all expunged, but if the same three felonies occurred between ages 19 and 24 (within a five-year time span) twenty years ago, only the most recent felony would be expungeable.

For Interpretive Approach 2 (counting *backward* from the date the application was filed), the court views the start of the applicable time period as the date the application for set aside was offically filed and counts backwards. This method is used by many courts and arguably justified by the Court's need to find the applicant is currently of good character and an asset to the community. In this approach, the 2022 misdemeanor would reset the wait period on all prior offenses, this is further complicated by the fact that the wait period for each offense varies. While the applicant

could get their 2022 misdemeanor set aside in 2025, since their 2008 Misdemeanor OWI carries a 5-year wait period they could not have it set aside until 2027. Additionally, their two felony convictions from 2006 and 2010 could not be set aside until 2029 because there is a 7-year wait period before multiple felonies can be set aside. In sum, this interpretation uses the filing date to start the look back period, and the length of the look back period is contingent on the type of offense that the applicant is seeking to have expunged. For example, if the applicant is seeking to have (a) Multiple felonies there is a7 Year look back from filing date, (b) 1 Felony or Serious Misdemeanors-5 year look back from filing date, and (c) for Non-Serious Misdemeanors there is a 3-year look back from filing date

More than three felonies and "One Bad Night". The statute can also be read to hold that a person with more than three felonies can be precluded from having any of their felony convictions set aside: "a person convicted of one or more offenses but not more than a total of three felony offenses... may apply to have all of his or her convictions... set aside". MCL 780.621(1)(a). This could mean that an upper limit of three convictions can be set aside, or it could mean that an individual with 4 convictions will not get any of them set aside.

In any event, the application process has a special exception for the all-too-common situation where several offenses occur as part of the same transaction within a 24-hour period. Consider a 2005 case where an applicant and a friend robbed a liquor store. They were convicted of five felonies: 1) breaking and entering, 2) theft of over $200 worth of merchandise, and when the police arrived, 3) fleeing and eluding law enforcement, 4) conspiring with others to do so, and 5) Operating While Intoxicated as they tried to get away. The fact that the applicant had 4 felonies in one day would ordinarily make them, and all of their convictions ineligible for expungement. MCL 780.621(1)a. But the statute's "One Bad Night" counting rule means that the court could count all these felonies as "a single felony" for the purpose of the statute. See MCL 780.621b(1). Thus, if there were no other incidents, these acts would all be expunged under the application process. Indeed, the applicant could have two more felonies and still have all of the offenses set aside.

5 Rules Engine to the rescue(?)

Acknowledging that the individual application process is time-consuming for the courts, and both difficult and expensive for the average citizen to pursue, the Michigan State Legislature directed the Michigan State Police (MSP) to create an automated process to relieve this pressure. (MCL 780.621g). The MSP collaborated with the State's Department of Technology, Management & Budget to implement and deploy an automated system — a "Rules Engine", in the MSP's parlance — which went into operation in April 2023 [19, 26]. In the first year the statute was in full effect and before any automated process was available, estimates are that out of about 2.8 million people with criminal records, the MSP estimates that the courts processed about 16,283 applications to set aside convictions [20]. In less than a month since automation was introduced in April 2023, it was estimated that 252,000 citizens have had their records cleared and 1.2 million offenses have been expunged [29].Thus, even though a wider variety of offenses are eligible for set-aside sooner under the application process, the automation process is currently generating the vast majority of expungements under the Clean Slate law.

The legislature intentionally created some limitations on what can be set aside under the automated process as compared to the judicial process. First, the wait time for automatic expungement is

longer (ten years rather than five to seven for felonies, and seven years rather than three for most misdemeanors). Moreover, there is an expanded list of non-expungeable offenses. A single OWI (Operating while Intoxicated) cannot be expunged automatically, and no "assaultive crimes" (for example, resisting arrest) nor "crimes of dishonesty" are eligible for automatic expungement. See MCL 780.621(g) (5-15).

The benefit of expunging convictions at the rate and scale facilitated by the Rules Engine is undeniable. Yet there are some aspects of the MSP's solution, as currently implemented, that deserve further scrutiny. First, it seems reasonable to assume that in a given case, what could be set aside automatically would be a simple subset of what a citizen would be eligible to see set aside by individual application. The automatic process created by the MSP, however, is both more restrictive and, in some aspects, more lenient than what courts have allowed for during the time before the Rules Engine. It is lenient in the sense that individuals who have had more recent convictions that occurred during their "look back period" will not be penalized in the manner that they would have been had they gone through the petition process. In theory, as long as their offenses don't overlap into the wait periods of the other offenses, the rules engine should auto expunge all eligible offenses. Yet the program can be more restrictive in nature, because any ineligible offense within a chain of offenses would block the rules engine from expunging prior offenses. As previously mentioned, most offenders generally accumulate convictions within a short period of time, thus the Rules Engine is more likely to encounter individuals who have multiple intervening convictions.

The rollout of the Rules Engine was an opportunity to provide a unified, clear message about the meaning of the statute, but so far that opportunity has not been taken.

Further, without more public education, resources, and transparency by the MSP as to how the rules engine works, most individuals who do benefit by getting some offenses set aside via the automatic process will never know that they could have other offenses set aside by application to the courts. Indeed, they may never know that they had any offenses set aside automatically, as they have to initiate that query and pay to generate an electronic copy of their record through IChat (Internet Criminal History Access Tool).

Of even greater concern is the risk that many, including attorneys and judges, will believe that since the automated process for set-asides have occurred, any remaining offenses are somehow more serious and either non-expungeable or should be more carefully scrutinized before additional set-asides are applied for or ordered.

Inside the Rules Engine. In creating the Rules Engine for the automatic set-aside process, the MSP had an opportunity to encode its own interpretation of the law that may have been based on ideological or pragmatic considerations rather than concerns about statutory interpretation or fealty to an interpretive methodology. Practitioners have apparently had little involvement or information as to how the MSP developed the Rules Engine, and there is no publicly available explanation of how it operates. Our description of the Rules Engine algorithm comes from a March 2023 MSP webinar for legal professionals and an accompanying PowerPoint document [20].

The presentation describes the operation of the Rules Engine as an iterative process, proceeding through an individual's conviction record in a series of passes, each pass from oldest to most recent. The Rules Engine is both forward-looking and backward-looking, as defined in §4, but with a difference: a conviction that is blocked from expungement by a following conviction within the

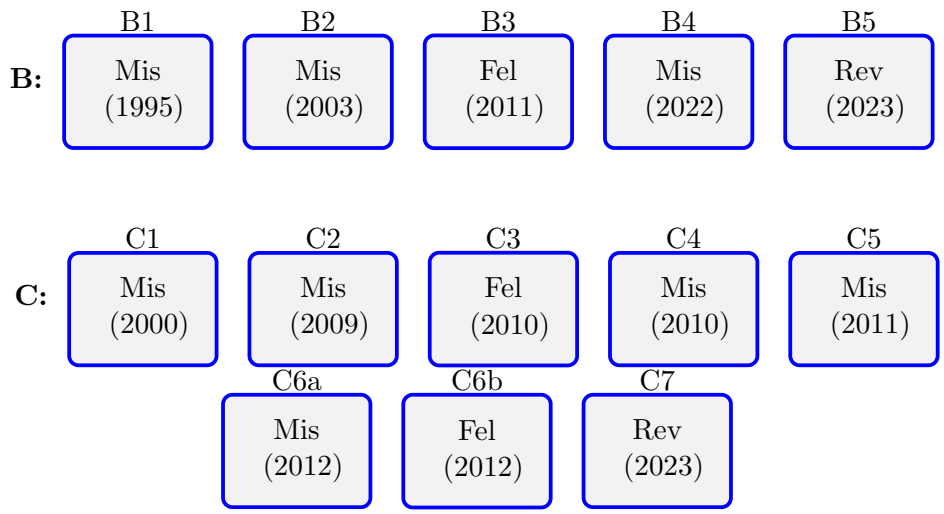

Figure 2: Expungement scenarios B and C.

"applicable time period" can eventually become expungeable if later iterations of the Rule Engine find the later conviction expungeable. The Engine first looks at the oldest offense, then looks forward for any offense within the applicable time period (for the Rules Engine, seven years for misdemeanors and ten years for felonies, which are longer waiting periods than what the statute stipulates). If none is found, the engine will set aside the oldest offense; otherwise, the process continues with the next offense in the individual's record. If Offense 2 is set aside, the process will eventually return to Offense 1 and, in the absence of any other remaining conviction in the applicable time period, will set it aside.

The algorithm at the core of the Rules Engine is explained in the Powerpoint Presentation offered by the MSP through two scenarios, shown in Fig. 2. In Scenario B, convictions 1 through 3 will be set aside by the rules engine on the first pass through the system. Conviction 4 will not be automatically set aside due to the backward looking stipulation, because the requisite seven years for misdemeanors have not passed yet. It will be eligible to be automatically set aside in 2029, provided the set aside is permissible at that time per MCL 780.621g.

In Scenario C, the order of setting aside via the rules engine would be as follows:

1. C1 is expunged, since the next offense (C2) occurred after the seven-year waiting period.

   C2-C6a are not expunged in the first pass, since for each of these offenses (looking forward) there is a following offense within the waiting period.

2. C6b is expunged, since (looking forward) there is no following offense within the waiting period and (looking backward) the appropriate waiting period has been reached by the time of the review.

   The expungement of C6b now removes this offense from consideration in further expungements, since according to MCL 780.622: "upon the automatic setting aside of a conviction under section 1g, the applicant... is considered not to have been previously convicted..."

3. This then allows C6a to be expunged in the second pass,

4. which allows C5 to be expunged in the third pass,
5. and similarly C4 in the fourth pass,
6. and C3 in the fifth pass.

In the end, only offense C2 due to the maximum allowance being reached. [20]

The presenters did not discuss explicitly how the Rules Engine would process a similar record that included offenses that were not automatically expungeable (*e.g.*, an OWI offense). According to the explanation of the process as offered in the presentation, such offenses could constitute serious obstacles for expungement. For instance, if offense C6a in Scenario C was an OWI (and therefore not automatically expungeable), its presence in the record would block all further expungements (of C3–C5). The rather substantial set of crimes that cannot be expunged automatically will surely have a significant impact on the expungement output of the Rules Engine. This has been borne out by the current results, where the estimated 400,000 citizens who could be conviction-free after April 11, 2023 has been reduced to 252,000 [20, 29].

6 Modeling and analysis with Alloy

Our investigation of Michigan's Clean Slate statute has revealed significant barriers for ordinary citizens, confusion even among legal experts, and unresolved differences in interpretation between lawyers and judges. Our goal is to increase participation, collaboration, and transparency among the stakeholders in Clean Slate, and ultimately other examples of statutory law. The research objective of this project is to develop a transformative, technology-empowered approach to inter-stakeholder communication, guiding stakeholders by formalizing statutory language, defining terms carefully and unambiguously, comparing different interpretations (such as the forward-looking and backward-looking interpretations of waiting periods described in §4) and anticipating the consequences of the proposed rules — including "black swan" scenarios [12] (such as the cascading effect of non-expungeable offenses described in §5).

The Rules Engine of the Michigan State Police provides a useful counterpoint. The Rules Engine is essentially a black-box interface to the Clean Slate statute, indicating which convictions may be expunged within a given scenario, according to a fixed interpretation (which happens to differ in important ways from the interpretations used by judges), but providing relatively little in terms of explanation. In contrast, we envision a digital sandbox for experimenting with statutory law — a tool that not only allows users to explore consequences of existing law under a set interpretation, but also compare the consequences of different interpretations, and even create and explore the consequences of extensions or amendments to existing law. In computer science, the metaphor of a sandbox is commonly used to suggest a safe place to explore the effects and implications of design choices before they are released into production.

To explore the potential of computational modeling for exploration and learning in the context of Clean Slate, we have been using the modeling language Alloy and the Alloy Analyzer to provide hands-on experience with automated feedback [10, 4]. An Alloy model captures real-world concepts and entities as abstractions, called signatures, and enables designers to define relations between signatures. Designers can then specify a set of mathematical constraints on signatures and relations. Constraints may be expressed in Alloy as first-order formulas, as relational expressions,

or a combination of both. In addition, the temporal operators of Linear Temporal Logic (LTL) are built into the language. The Alloy Analyzer searches within designer-specified bounds for instances satisfying an Alloy model, through a SAT-based constraint solver. If a satisfying instance is found, the analyzer displays it graphically; otherwise, it identifies a minimal unsatisfiable core of the model. Alloy has been used in a wide array of application areas, including security, computer architecture, program verification, and model driven development [11]. Our initial efforts, illustrated by the examples of §4, are described in some detail in Appendix A.

7 Related Work

There have been a number of related efforts, leveraging computation to broaden access and engagement in regulatory and legal contexts: for example, automated forms that facilitate legal services such as eviction help [25], and auto-generated insurance contracts that are compliant with legal rules [5]. The Blawx system offers a block-based programming environment for generating legal services such as drafting a will or creating a lease [21]. The creators of these projects argue that they lead to legislation that is clearer for all participants. Experiments with generating machine-readable regulations in New Zealand have indicated a speedup in interactions between government and business. Furthermore, machine-readable legislation allows stakeholders and public servants to test the impact of policy before they implement it [6]. Similarly, there are ongoing attempts to encode jurisprudence: for instance, court decisions that define and elucidate the regulatory environment [15]. The preliminary success of these efforts has inspired this proposed project.

There are numerous methods for the formalization of legal documents, most of which focus on developing specification languages for legal requirements, extracting legal requirements and model-based analysis of legal requirements. For example, frame-based methods provide logical roles and templates to classify legal phrases [2]. Breaux and Gordon [3] present a Legal Requirements Specification Language (LRSL) that is understandable for policy makers, legal experts and software developers. Their focus is specifically on specifying multi-jurisdictional legal requirements for distributed software so designers can analyze the consequences of implementing potentially conflicting regulations. This is achieved through formalizing the cross links between legal requirements specified in the original documents. Giorgini *et al.* [9] model requirements through ownership, permission and delegation. Maxwell and Antón [17] extract requirements from legal text using production rules. Massey and Antón [16] use some metrics to measure dependencies between regulations, which is one of the most important challenges in formalizing legal requirements. Some researchers use AI techniques for the formalization of legal regulations such as the representation of legal statutes as logic programs [27].

Soavi *et al.* [28] present a study on the formalization of business contracts specified in natural language. Based on their study, there are four main steps to formalizing legal texts: (1) identifying concepts in text and semantically annotating them; (2) identifying the relations between such concepts; (3) domain modeling of the concepts, and finally (4) generating formal specifications from domain models. One of the challenges we have in this work includes the complexity of the Clean State statute in terms of temporal constraints of expungement, and the order by which these constraints are considered when it comes to expunging convictions. The statute leaves a lot of room for misinterpretations. As such, formalization of the statute in a language like Alloy enables scope-based analysis of temporal and logical constraints under different interpretations.

8 Future work: Toward a digital sandbox for analysis and comparison of legal interpretations

A key feature of the sandbox concept is its accessibility to legal professionals and even ordinary citizens, without need for computer science expertise. Hence it is important to identify and implement a toolbox of predefined elements that users can use to build their own statutory regulations. Following Morris's example with Blawx [21], these could even be provided to the user as graphical units in a block-based language. To that end, it is important to identify common patterns within statutory law that we can capture with reusable code modules. The timing constraints in the statute can be classified into several categories of patterns, which are quantified in terms of the amount of time elapsed between events:

- absence of events in some future time interval (*e.g.*, no expungement occurs within at least x years from completing the sentence of a crime);
- absence of events in some past time interval (*e.g.*, if two crimes are committed in different occasions, then the second one can be expunged only if at least x years have past since the first one);
- length of time interval between events (*e.g.*, once a misdemeanor is served, it cannot be expunged until three years), and
- existence of an event between two other events (*e.g.*, Is there an intervening felony between an expunged felony and 7 years before it?)

Dwyer et al. [7] provide specification patterns that capture the qualitative versions of the aforementioned constraints in that there is no specific amount of time between the occurrence of events. For instance, the Existence pattern specifies whether some event P exists between two other events Q and R, but does not stipulate any constraint on how far Q and R should be from each other. As another example, the Precedence pattern requires that some event S occurs after another event T (*e.g.*, the expungement of a crime always occurs after its term is served). To this end, we plan to investigate the use of real-time specification patterns [13] in formal and visual specifications of timing constraints.

Another important question is the choice of the specification language. In this study, we have used Alloy due to its modeling power and automated scope-based analysis. However, we believe that no single formal language will be a silver bullet for the specification and analysis of the statutory language. To address this problem, we will study the use of other formal languages (*e.g.*, Metric Temporal Logic - MTL) in combination with Alloy. For example, MTL provides time-constrained versions of the temporal operators such as eventually, always and after. Such time-constrained operators will be very useful in exact specification of timing restrictions of the statute. Moreover, we plan to use visual specification languages to simplify the understandability of formal specifications for layman stakeholders. For instance, we will study how the Real-Time Graphical Interval Logic [22] can help us visually represent the timing constraints and other queries on formal models.

In sum, our ideal sandbox environment will allow differing interpretations of statutory law, to be expressed formally and compared through automated analysis. We see computation as a way not only to expand the benefits of criminal justice reform but also to enhance individuals' understanding and agency with regard to the law, thereby also enhancing stakeholders' ability to clearly articulate and advocate for needed change.

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A Alloy implementation

To explore the potential of computational modeling for exploration and learning in the context of Clean Slate, we use the modeling language Alloy and the Alloy Analyzer to provide hands-on experience with automated feedback [10, 4]. An Alloy model captures real-world concepts and entities as abstractions, called signatures, and enables designers to define relations between signatures. Designers can then specify a set of mathematical constraints on signatures and relations. Constraints may be expressed in Alloy as first-order formulas, as relational expressions, or a combination of both. In addition, the temporal operators of Linear Temporal Logic (LTL) are built into the language. The Alloy Analyzer searches within designer-specified bounds for instances satisfying an Alloy model, through a SAT-based constraint solver. If a satisfying instance is found, the analyzer displays it graphically; otherwise, it identifies a minimal unsatisfiable core of the model. Alloy has been used in a wide array of application areas, including security, computer architecture, program verification, and model driven development [11].

In our model, instances consist of temporal sequences of events (convictions and expungements), each with an associated date. `Event` and `Date` are defined as signatures in our model. Convictions are subclassed into misdemeanors and felonies, with further subclasses that are relevant to the language of the statute.

```
-- An event is a conviction or an expungement
abstract sig Event { date: one Date } -- each event has an associated date
-- A conviction is a felony or a misdemeanor
abstract sig Conviction extends Event { }
abstract sig Felony, Misdemeanor extends Conviction { }
-- Special types of felony: assaultive, ten-year
sig AssaultiveFelony, TenYearFelony in Felony { }
-- Special type of misdemeanor: OWI (Operating While Intoxicated)
sig OWI in Misdemeanor { }
-- Each expungement has a set of related convictions
-- note: multiple convictions may be expunged in a single event
sig Expungement extends Event { con: some Conviction }
```

At each point in time, a set of events may occur, represented by the variable signature `now`. We allow simultaneous convictions to capture the possibility of multiple criminal activities happening in "one bad night". Using Alloy's built-in temporal operators `eventually` and `after`, we define a happens-before relation over the `Event` signature, The `eventually` operator means that some condition holds now or at some future point of time, and `after` means that some condition holds in the next point of time. The predicate `hb` behaves as a Boolean function that returns true if the constraint inside it holds, and returns false otherwise. Note that `hb` is irreflexive; i.e. it holds for pairs of events `[e1, e2]` where `e1` strictly precedes `e2`.

```
-- now indicates the current event
var sig now in Event { }
-- The strict happens-before relation for events (no reflexive pairs)
pred hb[e1, e2: Event] { eventually (e1 in now and after eventually e2 in now) }
```

In order to capture the waiting-period aspects of the statute, we introduce a linearly ordered set of dates, with binary relations to capture the relevant waiting periods mentioned in the statute. For example, the binary relation `withinThree` captures all ordered pairs `d1->d2` where `d2` is within a three-year span of `d1`.

```
open util/ordering[Date]   -- Dates are linearly ordered
-- Associated with each date is the events occurring within 3/5/7 years of this date
sig Date { withinThree, withinFive, withinSeven: set Date }
```

We express the statute's restrictions on expungements as temporal logic formulas. For instance, Section 1,1b states that no more than two assaultive felonies may be expunged. We define a predicate to capture the occurrence of three (or more) assaultive felonies (using our previously defined happens-before relation `hb`) and then assert that in such cases the third assaultive felony is not expunged.

```
-- Does the assaultive felony af occur after two preceding assaultive felonies?
pred afterSecondAssault[af: AssaultiveFelony] {
   some disj af1, af2: AssaultiveFelony | hb[af1, af] and hb[af2, af]
}
-- No more than two assaultive felonies may be expunged (Sec. 1, 1b).
pred sec1_1b { no af: AssaultiveFelony | afterSecondAssault[af] and expunged[af] }
```

We can verify the expungeability of a given conviction by setting up a scenario that instantiates the relevant events and their relative time orderings. For instance, consider a simple scenario where expungements are sought for two assaultive felony convictions. We define a predicate capturing the details of the scenario and instructing the Alloy Analyzer to find a satisfying instance:

```
pred show {
   some af1, af2: AssaultiveFelony |
      hb[af1, af2] -- af1 happens first
      and af1.expunged and af2.expunged -- both felonies are expunged
}
run show for 4 Event, 3 Date
```

Note that we can specify upper bounds on the sizes of the signatures `Event` and `Date`. In this case, there is a satisfying instance (in fact, many satisfying instances) with only four distinct events and three distinct dates. The Analyzer gives a visualization of the instance as a sequence of states, with the initial state displayed in Fig. 3.

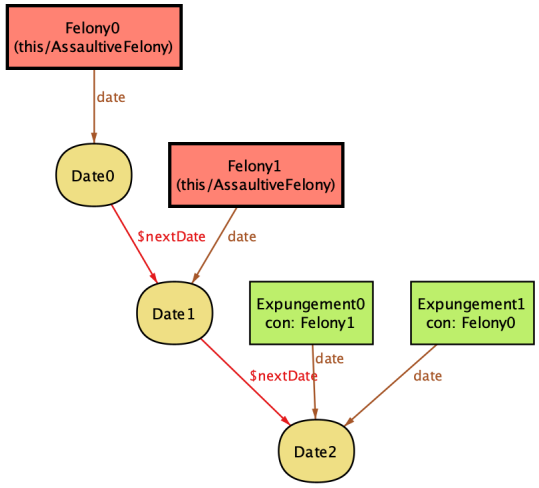

Figure 3: Two assaultive felonies expunged.

Although the instance involves three states, the initial state actually provides all the information needed to understand the scenario, due to the timestamping provided by the `date` relation. Here, `Felony0` occurs first, followed by `Felony1` (more than seven years later), followed (more than seven years later) by an expungement of both `Felony0` and `Felony2`.

If we attempt to extend the scenario with a third expunged assaultive felony, Alloy responds with a failure message and an indication that the scenario is incompatible with the predicate `sec1_1b`:

```
pred show {
   some af1, af2, af3: AssaultiveFelony |
      hb[af1, af2] and hb[af2, af3]
      and af1.expunged and af2.expunged and af3.expunged
}
run show for 6 Event, 6 Date
Executing "Run show for 6 Event, 6 Date"
No instance found. Predicate may be inconsistent.
```

In addition to simple expungement testing for specific scenarios, the Alloy analyzer allows us to explore other questions regarding interpretations of the Clean Slate Act. For instance, we can express the difference between the "backward" and "forward" interpretations of waiting periods, described earlier, in terms of different predicates.

```
pred backwardWaiting {
   no x: Expungement |
    (some m: Misdemeanor | x.date in m.date.withinThree)
    or (some f: Felony | x.date in f.date.withinFive)
    or (some f: Felony | interveningFelony[f, x] and x.date in f.date.withinSeven)
}
```

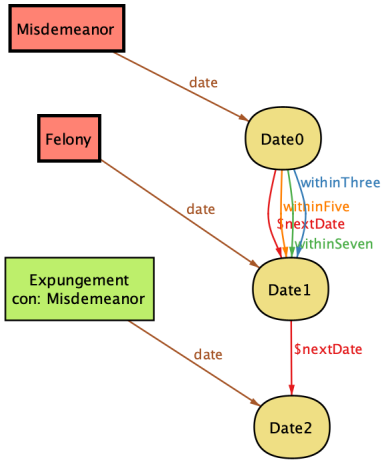

Figure 4: Two assaultive felonies expunged.

```
pred forwardWaiting {
   all m: Misdemeanor |
      m.expunged implies no c: Conviction - m | c.date in m.date.withinThree
   all f: Felony |
      f.expunged implies no c: Conviction - f | c.date in f.date.withinFive
   all f1: Felony |
      f1.expunged implies no f2: Felony - f1 | f2.date in f1.date.withinSeven
}
```

The `backwardWaiting` predicate asserts that for any expungement, there is no preceding conviction in the "waiting period" (three years for a misdemeanor, five years for a single felony, and seven years for a felony with additional intervening felony). In contrast, the `forwardWaiting` predicate asserts that for any conviction, there is no *following* conviction in the waiting period.

We can view the differences between these two interpretations by instructing the Alloy Analyzer to find instances that satisfy one predicate but not the other.

```
pred show {
   backwardWaiting
   not forwardWaiting
}
run show for 3 Event, 3 Date
```

This will search for instances satisfying the backward interpretation of waiting periods but not the forward waiting interpretation. Note that upper bounds for the sizes of the `Event` and `Date` signatures are given; here we can find an instance with only three distinct events and three distinct dates. The Analyzer indicates the existence of a satisfying instance and provides a visualization, shown in Fig. 4.

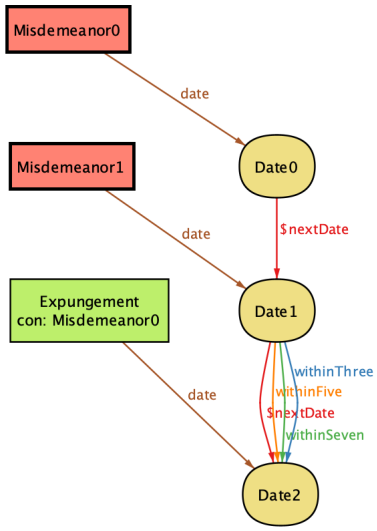

Figure 5: Two assaultive felonies expunged.

The `Expungement` at `Date2` expunges the `Misdemeanor` at `Date0`. Note that `Date1` is within three years of `Date0`, but `Date2` is more than seven years after `Date1`.

This is indeed compatible with the backward interpretation of waiting periods, since `Expungement` takes place more than seven years after the previous `Felony` conviction. (In fact, under the backward interpretation, only three years since `Felony` would be sufficient.) On the other hand, it is not compatible with the forward interpretation, since Misdemeanor is followed by another conviction, `Felony`, before the waiting period of three years has expired.

Alternatively, we can search for instances satisfying the forward waiting interpretation but not the backward waiting interpretation.

```
pred show {
   not backwardWaiting
   forwardWaiting
}
run show for 3 Event, 3 Date
```

The Analyzer indicates the existence of a satisfying instance and provides a visualization, shown in Fig. 5.

In this instance, `Misdemeanor1` takes place more than seven years after `Misdemeanor0`, so it does not constitute an obstacle to expungement of `Misdemeanor0` under the forward-looking interpretation. But `Expungement` does occur within three years of `Misdemeanor1`, which runs contrary to the backward-looking interpretation.

We can also use the Alloy Analyzer to clear up misconceptions regarding the consequences of the statute. For instance, it may be surprising to discover that an individual may have more than three

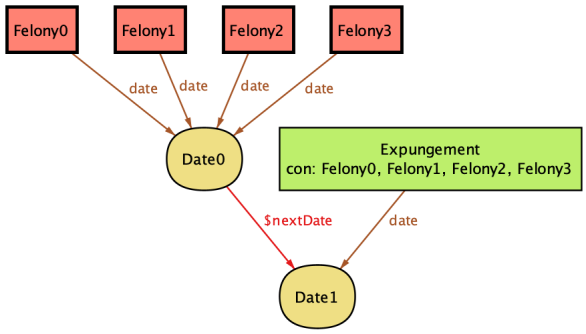

Figure 6: Two assaultive felonies expunged.

felonies expunged. At first reading, this appears to contradict Section 1,1a of the statute, which states, "No conviction may be expunged after three or more felonies". The relevant code from our Alloy model is given below.

```
-- Does conviction c occur after three preceding felonies?
pred afterThirdFelony[c: Conviction] {
   some f1, f2, f3: Felony | hb[f1, f2] and hb[f2, f3] and hb[f3, c]
}
-- No conviction may be expunged after three or more felonies (Sec. 1, 1a).
pred sec1_1a { no c: Conviction | afterThirdFelony[c] and expunged[c] }
```

The key insight in our model is that multiple events can happen simultaneously, but since the `hb` relation is irreflexive, the `afterThirdFelony` is only counting sequences of felonies that happen at different times. This is compatible with the One Bad Night rule, which intentionally allows for multiple offenses that occurred at the same time to be counted as one offense for Section 1,1a and other clauses that "count" offenses.

We can instruct the Alloy Analyzer to search for instances with four expunged felonies.

```
some disj f1, f2, f3, f4: Felony |
   f1.expunged and f2.expunged and f3.expunged and f4.expunged
```

The Analyzer finds an instance involving a single "bad night" with four felonies, all of which are expunged at the same time (Fig. 6).